

Summer Semester**MID-SEMESTER THEORY EXAMINATION, June, 2023**Course Code- EEEEC03/~~EEEC03~~ 1C1CC03 / ECECC05

Course title- Signal and System

Time- 1:30 hour

Max. Marks- 25

Note: Attempt all questions. Missing data (if any) may be suitably assumed and mentioned in the answer sheet

Q.No	Questions	Marks	CO
1a	Define signals and explain continuous time and discrete time signals with the help of diagram	2.5	CO1
1b	Discuss different operations on continuous time signal (by using unit step function $u(t)$; Time delay/advance, Time folding, Time scaling	2.5	CO1
2a	What are even and odd signals, and prove that any signal can be represented as the components of even and odd signal?	2.5	CO1
2b	Determine signal $f(t)$ is even or odd? $f(t) = t^3 \sin t$	2.5	CO2
3a	Define unit step function and draw the waveform for the signal $V(t) = u(t-1) + u(t-2) + u(t-3) + u(t-4) - 4u(t-5)$	2.5	CO1
3b	The unit step response of a system is $(1 - e^{-at}) \cdot u(t)$, then its impulse response is?	2.5	CO2
4a	Determine the Laplace transform of the signal $F(t)$, $F(t) = 10.t e^{-2t}$	2.5	CO2
4b	The continuous-time system defined by $Y(t) = x(t^2)$; Comment whether the system is linear, causal and time in-variant or not ?	2.5	CO1
5a	Define convolution for a linear time invariant (LTI) system, and discuss the steps involved in determining convolution of an LTI system?	2.5	CO2
5b	Find the output $y(t) = ?$ for the given continuous time LTI system with unit impulse response $h(t)$. $x(t) = e^{2t}u(t)$ $h(t) = u(t-3)$	2.5	CO1